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The Effects of Non –Surgical Periodontal Therapy on Glycemic Control in Diabetic Patients (Randomized Controlled Trial)

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Abstract

Aim: The present study aimed to monitor the clinical outcomes and the metabolic response of periodontal therapy (non-surgical) in patients with periodontitis (chronic) and uncontrolled diabetes (type 2).

Methods: Eighty-eight subjects with periodontitis (chronic) and uncontrolled diabetes (type 2) were enrolled in this controlled trial and allocated randomly to test group (44 patients were received immediate periodontal therapy) or control group (44 patients were received delayed periodontal therapy). The metabolic and clinical evaluations were conducted at baseline and 3 months. This included clinical attachment level, Glycated hemoglobin (HbA1c), bleeding on probing, visible plaque, and pocket depth. The periodontal therapy in this study consists of one stage scaling and root planning, a combination of systemic antibiotics (amoxicillin 500mg and metronidazole 400 mg), and oral hygiene instructions.

Results: Regarding clinical and metabolic parameters at baseline, no statistically significant differences were displayed between two groups. However, at 3-months follow-up period the patients within a test group demonstrated significantly better clinical and metabolic outcomes than patients in a control group.

Conclusion: The non-surgical periodontal treatment using a combination of metronidazole and amoxicillin significantly improved the metabolic outcome in addition to periodontal health in diabetic subjects with chronic periodontitis.

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Clinical relevance:

Scientific rationale*:* The existing evidence on the link between glycemic control and periodontal therapy remains unclear, in spite of several scientific articles that studied such link.

Principle findings: Periodontal therapy significantly improve both glycemic control and periodontal health in diabetic patients.

Practical implications: As the non-surgical periodontal therapy can significantly improves glycemic control in diabetic patients, dental consultation should be an important part during the management of diabetic patients.

Introduction

Periodontitis is a chronic multifactorial infectious and inflammatory disorder manifested by the loss of structures that support teeth(Pihlstrom, Michalowicz, & Johnson, 2005). The epidemiological studies reported that the prevalence of periodontitis is 15-30% of adult populations(König, Holtfreter, & Kocher, 2010). Clinical manifestations of periodontitis include increased clinical attachment loss or pocket depth. Periodontal therapies consist of instructions for oral hygiene, scaling, root planing, adjunctive use of antibiotics and or chlorhexidine, surgical therapy, or a combination of these (Armitage, 2004). Miranda et al., (2014) reported that in the treatment of Diabetes mellitus type 2 (T2DM) with chronic periodontitis the adjunctive use of amoxicillin and metronidazole significantly improved the microbiological and the clinical results. The prevalence of diabetes among adults from 20 to 79 years in Egypt is around 15.56%.(Hegazi, Elgamal, Abdel-Hady, & Hamdy, 2015). T2DM caused by defects in insulin activity, production or both can cause macro and microvascular complications. Therefore, adequate glycemic control is critical to prevent these complications (Yamaguchi & Sugimoto, 2011). The common test used in patients with diagnosed diabetes to check glycemic control is Glycated hemoglobin (HbA1c) (American Diabetes Association, 2016). HbA1c shows serum glucose levels within the previous 120-days period and is considered a robust indicator for monitoring glycemic control (Higgins, 2013). A link exists between the development of complications and increasing HbA1c levels (Rohlfing et al., 2002). In the past decades, the relationship between chronic periodontitis and diabetes has been the aim of the most intensive research (Monsarrat et al., 2016). Many studies suggested the concept of

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a two-way relationship between both diseases (Taylor, Preshaw, & Lalla, 2013). Type 2 diabetic patients show a higher extent, severity, and prevalence of chronic periodontitis than healthy subjects (Chávarry, Vettore, Sansone, & Sheiham, 2009). Accordingly, T2DM is considered a risk factor for chronic periodontitis (Jimenez, Hu, Marino, Li, & Joshipura, 2012). Simpson et al .,(2010) observed that the deterioration of glycemic control attributed to chronic periodontitis over time(Simpson, Needleman, Wild, Moles, & Mills, 2010). Periodontal therapy can decrease local inflammation, the systemic entry of pro-inflammatory molecules and bacterial by-products such as lipopolysaccharides (Taylor et al., 2013). Simpson et al .,(2015) in a Cochrane systematic review, reported that many researches have studied the relationship between glycemic control and periodontal therapy (non-surgical modality) in the diabetic population and have demonstrated contradictory findings(Simpson et al., 2015). Several clinical trials recommended that more studies are necessary to clarify this link (Janket, Wightman, Baird, Van Dyke, & Jones, 2005); (Simpson et al., 2010); (Teshome & Yitayeh, 2017). Therefore, this clinical trial aimed to study the influence of periodontal therapy (non-surgical) on the glycemic level among T2DM patients with chronic periodontitis.

Methods and Materials

The protocol of the current trial was approved by Taibah University, Research Ethics Committee (TUCDREC/20181205/Makaky). This trial was carried out according to guidelines of the declaration of Helsinki. The ClinicalTrials.gov identifier of this trial is NCT03783845. The current study was followed the CONSORT (version 2010) guiding principles.

Sample size computation

HbA1c as a primary outcome was used to compute sample size. The difference in HbA1c between test and control groups was found to be 0.3% (standard deviation of 1.6%) at 3 months in a previous trial (Rodrigues, Taba Jr, Novaes Jr, Souza, & Grisi, 2003). Depending on these data, the ideal sample size for each group was 42 patients to guarantee power of 80% with two-sided confidence level of 95%. The total sample size was 88 patients to compensate for the dropout rate throughout the study.

Study groups and design

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A prospective, single-blinded, randomized controlled clinical trial of 88 patients aged 40-70 years was conducted on two parallel groups with three months follow up period.

Test group (patients were given immediate periodontal therapy)

A control group (patients were given delayed periodontal therapy)

The subjects with uncontrolled type 2 diabetes (HbA1c from 7% to 9%) were selected from attendants of Internal Medicine Outpatients Clinic, Tanta University Hospital between Jun 2015 and March 2016, and were referred to Periodontology Department, Tanta University.

A code number was used for each patient, and data were recorded in a medical archive created specifically for this trial and saved in a computer database. Closed envelopes were used by the study coordinator (SH) to randomly allocate the patients to a test and controlled group, using a 1:1 allocation ratio. This random series was hidden from the principal investigator (MY) who screened the patients. The same periodontics specialist (MY) treated all patients in both groups. The clinical parameters in both studied groups were recorded by the same examiner (SH) who was blinded to metabolic parameter data and the intervention protocol. Laboratory technicians who carried out the HbA1c analysis were blinded to the test group assignments via using patient code only to identify each vial. All patients signed informed consent.

Inclusion criteria

- Diagnosis of diabetes type 2 for more than five years
- Type 2 diabetes with HbA1c level from 7% to 9% at the last medical evaluation
- Age from 40 to 70 years
- No change in diabetes treatment over the previous three months
- Minimum sixth permanent teeth excluding third molars
- Clinical attachment level (CAL) and pocket depth (PD) ≥4 mm in more than thirty percent of the sites
- Diagnosis with chronic periodontitis
- Diagnosis of periodontitis based on the presence of 4 teeth as a minimum with at least one site with a CAL ≥ 3 mm and PPD ≥ 4 mm.

Exclusion criteria

- Pregnancy
- Alcoholism and smoking

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- Presence of any systemic disorders other than hypertension and diabetes
- Diabetic major complications
- Antimicrobial therapies in the last 6 months
- Periodontal therapies in the last 6 months
- Allergy to metronidazole and amoxicillin

Intervention protocol

Periodontal therapy included scaling and root planing (SRP), the combination of systemic antibiotics[metronidazole 400 mg three times daily for two weeks and amoxicillin 500mg three times daily for two weeks](Miranda et al., 2014) and oral hygiene instructions (OHI). Full-mouth SRP was performed by the same specialist(MY) for all patients in a test group using ultrasonic apparatus (Cavitron Select SPC, Dentsply professional, New York, USA) and Gracey curettes (Hu-Friedy, Chicago, IL, USA) in a single two-hour session. Local anesthesia was applied when needed. SRP was associated with subgingival irrigation by chlorhexidine 0.12% mouthwash (Peridex oral rinse, 3M ESPE Dental Products, U.S.A). Immediately, after SRP, the administration of antibiotics combination was started and patients received free dental care products necessary for periodontal maintenance during a follow-up period (three months). For the patients within a control group delayed periodontal therapy was performed using the same procedures that were applied for the patients in a test group exactly after the end of follow up period (three months) because of ethical reasons. Therefore, within the control group, no periodontal therapies were done during the follow-up.

Clinical and metabolic follow-up

A complete medical history, socio-demographic data, and information regarding daily dental care were collected for all patients. Charting for the full-mouth periodontium using a periodontal probe (Hu-Friedy, Chicago, IL, USA) was performed by the same specialist (SH). This includes PD (linear distance from margin of gingiva to base of pocket), CAL (the linear measurement from the cement–enamel junction to the base of the pocket), bleeding on probing (monitored as the percentage of areas that bleeds after probing by 15 seconds) and the visible dental plaque that assessed by passing a periodontal probe at four sites (buccal, lingual mesial and distal) for each tooth and recorded as (yes/no) (Quintero et al., 2018). The bleeding on probing (BOP), CAL and PD were recorded for all teeth at six sites for each tooth after third molars exclusion. The periodontal evaluation was done at baseline and three months. Test–retest exercises were done in 20 subjects before initiating the trial to assess intra-examiner calibration. The intra-examiner reproducibility for PD were 85%

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and for CAL were 80%. A venous Blood sample was taken from a median cubital vein to monitor HbA1c using high-performance liquid chromatography (Tosoh Bioscience, CA, USA) at baseline and three months post-treatment. The changes in HbA1c between the studied groups throughout follow-up periods have been considered as a primary outcome in this study. Secondary outcomes were changes in periodontal parameters between test and control groups throughout the study period.

Statistical analysis

Data were coded, collected and analyzed using the SPSS software under Mac OS Majave Version 10.14. Descriptive statistics were done using mean and standard deviations, followed by inferential statistics using the Fisher's exact test, Chi-Square test, student t- test and paired t-test. Significance was adopted at $p<0.05$ which used as a cutoff point to control for alpha error. For this study, intension to treat analysis was considered.

Results

None of the patients in a test group reported significant side effects after periodontal therapy. During the study follow-up period, no significant changes were monitored in physical activity and diet among the study population. All patients in both studied groups attended until the end of the study period (3 months). The socio-demographic data of the two groups were demonstrated in Table 1. The mean age in the test and control group was 52.95±6.523and 52.23±7.028, respectively without significant difference. The majority of the studied subjects were female 56.8%, within the test group the percent of a female was 59.1% while among the control group females represented 54.5%. However, no significant difference was found between the studied groups concerning sex. More than half of the studied sample had a family size of ≥ six members (61.4%) with no significant difference between studied groups regarding family size.

Those with secondary education represented 56.8% of the studied patients (50% in the test group and 63.7% within control group) while primary education ranked second representing 22.7% of the studied sample, the percentage was only 13.6% among control group compared with 31.8% among test group. However, the difference in educational levels between the different groups was not significant. Those of rural residence represented 50% of studied subjects with the highest percentage among the test group (54.5%). Differences between studied groups concerning the residence were found not statistically significant. The majority of the

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sample had a monthly income of ≤ 2000 Egyptian pounds (70.5%). This represented 72.7% of the test group and 68.2% of the control group with no statistical differences between the studied groups regarding monthly income.

It was found that at baseline there were no significant differences between studied groups regarding daily brushing and flossing while at 3 months these differences became significant (table 2). Within the test group, the mean value of brushing frequency and flossing were significantly increased with the time of recording from baseline to 3 months (table 3). While within a control group no significant changes were recorded from baseline to 3 months regarding brushing and flossing (table 4).

Within the test group, the mean values of visible dental plaque, PD, BOP, and CAL were statistically significantly decreased from baseline to 3 months postoperative (table 3). While among a control group the mean values of visible dental plaque, PD, BOP, and CAL were significantly increased from baseline to end of study (table 4). Besides, the differences between the studied groups at 3 months regarding these clinical variables were statistically significant (table 2).

The mean value of HbA1c for the test group was significantly decreased from baseline to 3 months. However, within a control group, it was significantly increased from baseline to termination of trial and the differences between studied groups at 3 months about the mean value of HbA1c were found statistically significant (table 2).

Discussion

Both periodontitis and diabetes are multifactorial and complex disorders. So determining the therapeutic impact of one on the other by clinical studies is not easy (Sadeghi, Taleghani, & Farhadi, 2014).In the present three- month controlled clinical study, an important aspect was a creation of an optimal non-surgical periodontal therapy (NSPT) (consist of OHI, SRP, combination of antibiotics, and antiseptics), versus no active therapy in control group, to enhance the probability of detection any impact of periodontal therapy on the HbA1C level. In the field, this maximization is unique because previous clinical trials either provided potentially effective periodontal therapy for control groups such as study of Engebretson et al., (2013) who stressed on oral hygiene instructions for control patients(Engebretson et al., 2013) or provided sub-optimal periodontal therapy such as study by Gay et al., (2014) who studied the impact of periodontal treatment composed of OHI and SRP alone(Gay et al., 2014). In this clinical study plaque indicator was monitored at

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different follow-up visits this is in agreement with Mizuno et al., (2017) who studied the impact of NSPT on oxidative stress balance, and glycemic level in T2DM patients and reported that plaque indicator is important to evaluate the patient's compliance with OHI(Mizuno et al., 2017). Following Madianos and Koromantzos, (2018) in the current study decreased HbA1C level was used as an indicator for benefits of periodontal therapy on a glycemic level within diabetes(Madianos & Koromantzos, 2018).

For diabetic subjects, the more efficient approach of NSPT to improve the periodontal health or the impact on glycemic level remains unclear (V. Santos et al., 2012; Vanessa R Santos et al., 2013). The traditional modality of NSPT, which composed of multiple visits of SRP only or in combination with adjunctive antibiotics and or antiseptics at 7-to 21-days intervals, has associated with clinical improvements in diabetic patients (Vanessa Renata Santos et al., 2009). The other short-term approach composed of full-mouth SRP within 24 hours, alone or with the use of chlorhexidine antiseptic (Apatzidou, Riggio, & Kinane, 2004).This one-stage approach has been established to prevent the translocation of microbes from untreated periodontal sites to treated ones and has shown microbiological and clinical results similar or better than those of the traditional modality (Quirynen et al., 2000), without additional risk in comparison to the traditional modality for the probability of microbial resistance development or patient's health (Teughels, Dekeyser, Van Essche, & Quirynen, 2009). Therefore, in the present clinical study, full-mouth SRP was done in a single two-hour session. In the present study, the antibiotic was prescribed. This coincided with Vergnes et al., (2018) that recommended the use of antibiotic therapy for uncontrolled diabetes to prevent local infection after invasive dental therapies(Vergnes et al., 2018). Furthermore, the data from the present study demonstrated that the use of a combination of amoxicillin and metronidazole as adjunctive to SRP yielded considerable clinical and metabolic advantages at 3 months follow-up period. This is in accordance with Souto et al., (2018) who reported that nonsurgical periodontal treatment using a combination of amoxicillin and metronidazole in diabetic patients associated with best clinical results(Souto et al., 2018).

The current trial studied the periodontal outcomes and the metabolic response of one-stage NSPT in subjects with chronic periodontitis and uncontrolled type 2 diabetes (T2DM). The data from this study demonstrated that this NSPT modality was effective and statistically significant improved the periodontal status in diabetic patients regarding CAL, BOP, PD, visible dental plaque and hygiene instructions. This is following Koromantzos et al.,(20011) and Santos et al., (2012, 2013) who observed that in diabetic patients the

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improvement in the periodontal status can be obtained using either one-stage or multi-session NSPT(Koromantzos et al., 2011). In diabetic patients the data on the influence of periodontal treatment on metabolic control are contradictory. Some publications have reported that NSPT may improve metabolic control (Madden et al., 2008), (Simpson et al., 2010), (Teeuw, Gerdes, & Loos, 2010), (Koromantzos et al., 2011), (Sgolastra, Severino, Pietropaoli, Gatto, & Monaco, 2013). However, in these patients, other studies cannot demonstrate significant improvement of glycemic control(Janket et al., 2005), (Jones et al., 2007), (Correa et al., 2010), (V. Santos et al., 2012), (Simpson et al., 2015).In T2DM patients, glycemic control is important to prevent complications. HbA1c is a crucial indicator of the glycemic level (Rohlfing et al., 2002). Stratton et al., (2000) who studied the impact of glycaemia on microvascular and macrovascular complications in T2DM patients, and concluded that each 1% decreasing of HbA1c value led to a 21% reduction in the relative risk of deaths due to diabetes, a 37% decreasing in the risk of microvascular complications, and a 14% reduction in the risk of myocardial infarction(Stratton et al., 2000). The current study demonstrated that NSPT can significantly reduce the level of HbA1c in T2DM patients with chronic periodontitis. This is in agreement with Quintero et al., (2018) who studied the impact of two NSPT modalities in T2DM patients with chronic periodontitis and reported that better periodontal clinical parameter and HbA1c levels improvements were achieved with both NSPT modalities. In the present study, the explanation for these results may be attributed to the elevated serum level of IL-6, IL-1 and TNF-a that associated with chronic periodontitis, could antagonize the action of insulin (Calle & Fernandez, 2012)and increase the insulin resistance (Tunes, Foss-Freitas, & Nogueira-Filho, 2010). Data from this clinical study stressed the importance of dental referral for T2DM patients during diabetes management. This is in accordance with Sadeghi et al., (2014) who studied the quality of life and oral health among diabetic patients and reported that referral of diabetic patients to a dentist is critical and should put as integral part of a diabetes treatment plan.

Limitations: Some limitations should be considered during the analysis of data obtained from this study, such as the follow-up period and sample size. Therefore, the future randomized controlled clinical trials with a longer duration and a larger population size should be formulated to validate the data reported here.

Conclusions

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The outcomes of this trial concluded that one-stage NSPT with a combination of amoxicillin and metronidazole significantly improved metabolic and periodontal outcomes in the management of chronic periodontitis in T2DM patients

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Tables

Table 1: Socio-demographic characteristics of studied groups

Character	Test group (n=44)		Control group (n=44)		Total (n=88)		Test	P-value
	n	%	n	%	n	%		
Age in years: mean SD	52.95 ±6.523		52.23 ±7.028				t-test = 0.503	0.616
Sex:								
Males	18	40.9	20	45.5	38	43.2	F =0.185	0.830
Females	26	59.1	24	54.5	50	56.8		
Family size:								
≤5	16	36.4	18	40.9	34	38.6	F =0.192	0.827
≥6	28	63.6	26	59.1	54	61.4		
Education:								
Illiterate	4	9.1	4	9.1	8	9.1		
Primary	14	31.8	6	13.6	20	22.7	X^2 =4.320	0.229
Secondary	22	50	28	63.7	50	56.8		
High	4	9.1	6	13.6	10	11.4		
Residence:								
Urban	20	45.5	24	54.5	44	50	F =0.727	0.523
Rural	24	54.5	20	45.5	44	50		
Monthly income:								
≤ 2000 EP	32	72.7	30	68.2	62	70.5	F =0.218	0.816
$> 2000\ EP$	12	27.3	14	31.8	26	29.5		

Standard deviation (SD) **Chi-square (X^2)** **p< 0.05 (Significant)** **Fisher's exact test (F)**

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Table 2: Comparison of clinical and metabolic outcomes among the studied groups

variable	Follow-up period	Test (n=44) X SD	Control (n=44) X SD	t-test	p-value	95% confidence interval
Brushing	Baseline	0.91±0.86	0.91±0.96	0.000	1.000	-0.39 to 0.39
	3-months	2.09±0.80	0.82±0.84	7.258	< 0.001	0.92 to 1.62
Flossing	Baseline	0.27±0.55	0.32±0.64	-0.359	0.720	-0.3 to 0.21
	3-months	1.50±0.67	0.32±0.64	8.503	< 0.001	0.91 to 1.46
Pocket depth (mm)	Baseline	4.99±0.75	5.01±0.83	-0.163	0.871	-0.37 to 0.31
	3-months	4.27±0.88	5.1±0.83	-4.534	< 0.001	-1.18 to -.46
Clinical attachment level (mm)	Baseline	5.27±0.80	5.24±0.82	0.185	0.854	-0.31 to 0.38
	3-months	4.47±0.73	5.41±0.81	-5.780	< 0.001	-1.28 to -0.63
Bleeding on probing(% sites)	Baseline	45.64±28.34	46.27±20.73	-0.120	0.905	-11.16 to 9.89
	3-months	15.36±9.82	49.23±20.75	-9.786	< 0.001	-40.74 to -26.99
Visible dental plaque	Baseline	45.77±28.1	46.41±20.69	-0.121	0.904	-11.09 to 9.82
	3-months	16.64±10.21	49.50±21.02	-9.329	< 0.001	-39.87 to -25.86
Glycated hemoglobin	Baseline	8.12±0.74	8.21±0.71	-0.443	0.659	-0.37 to 0.24
	3-months	7.27±0.5	8.34±0.64	-8.763	< 0.001	-1.32 to -0.83

X=Mean p< 0.05 (Significant) Standard deviation (SD)

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Table 3: Comparison of the outcomes within a test group at baseline – 3 months

variable	Standard deviation	t-test	p-value	95% confidence interval
Brushing	±0.657	-11.937	< 0.001	-1.381 to -0.982
Flossing	±0.803	-10.138	< 0.001	-1.471 to -0.983
PD Pocket depth (mm)	±0.445	10.711	< 0.001	0.583 to 0.854
CAL Clinical attachment level (mm)	±0.539	9.899	< 0.001	0.641 to 0.969
Bleeding on probing(% sites)	±19.321	10.393	< 0.001	24.398 to 36.147
Visible dental plaque	±19.396	9.964	< 0.001	23.239 to 35.033
Glycated hemoglobin	±0.532	10.763	< 0.001	0.702 to 1.026

p< 0.05 (Significant)

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Table 4: Comparison of the variables within the control group at baseline – 3 months

variable	Standard deviation	t-test	p-value	95% confidence interval
Brushing	±0.291	2.074	0.044	0.002 to 0.179
Flossing	t- test and correlation cannot be computed because the standard error of the difference is 0			
PD Pocket depth (mm)	±0.106	-4.865	< 0.001	-0.109 to -0.045
CAL Clinical attachment level (mm)	±0.087	-13.133	< 0.001	-0.993 to -0.146
Bleeding on probing(% sites)	±2.996	-6.542	< 0.001	-3.865 to -2.044
Visible dental plaque	±3.395	-6.039	< 0.001	-4.123 to -2.059
Glycated hemoglobin	±0.272	-3.325	0.002	-0.219 to -0.054

p< 0.05 (Significant)

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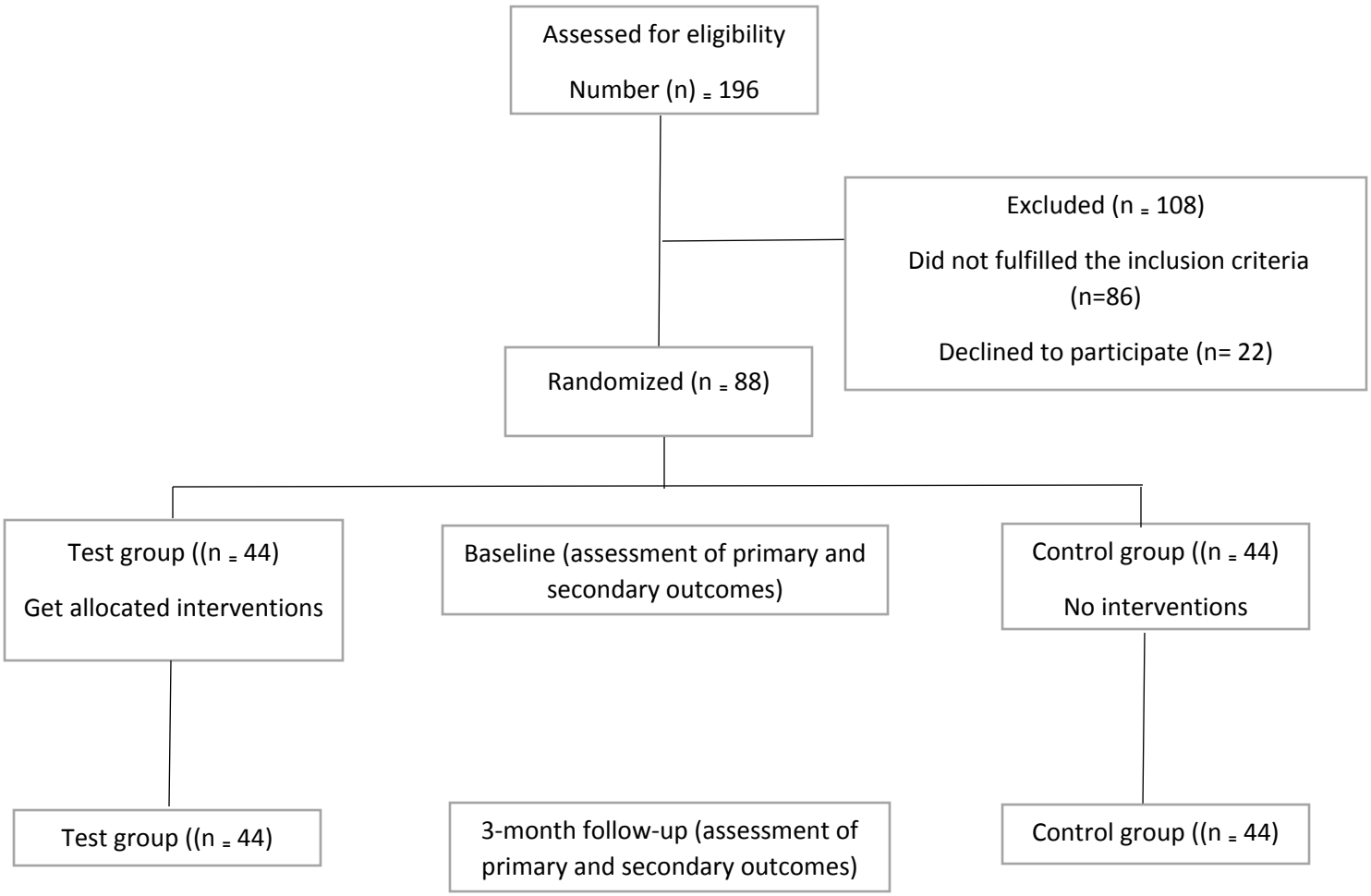

Flow diagram of enrolment of study patients

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